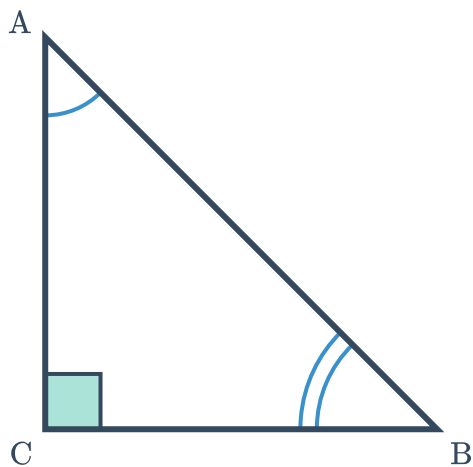


Right-angled triangles

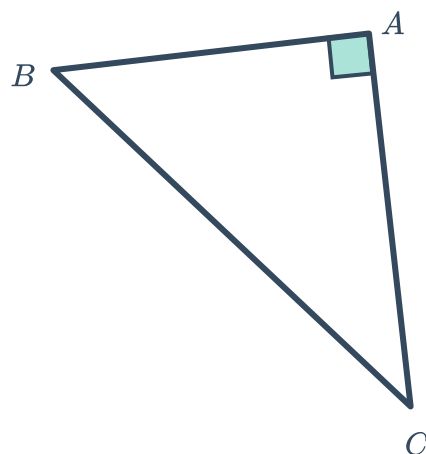


1 For the following triangles, name the hypotenuse:

a

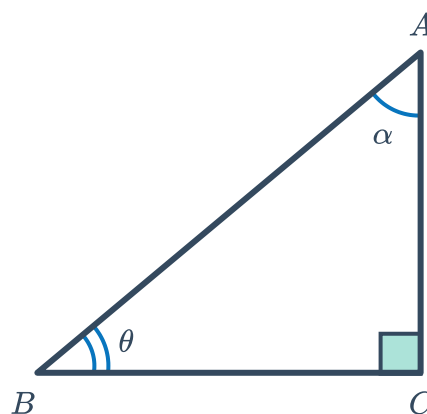


b



2 Consider the following triangle:

- a State the opposite side to angle θ .
- b State the adjacent side to angle θ .
- c State the opposite side to angle α .
- d State the adjacent side to angle α .
- e State the angle that is opposite the hypotenuse.



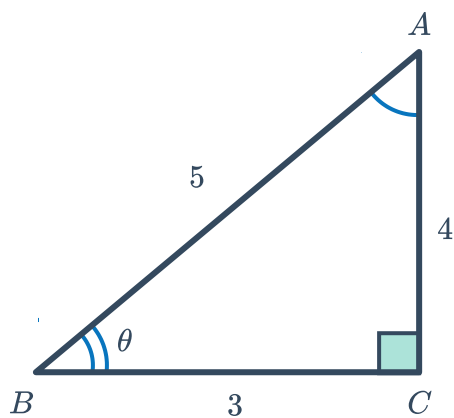
3 With reference the angle θ , find the value of these ratios for each of the following triangles:

i $\frac{\text{Opposite}}{\text{Adjacent}}$

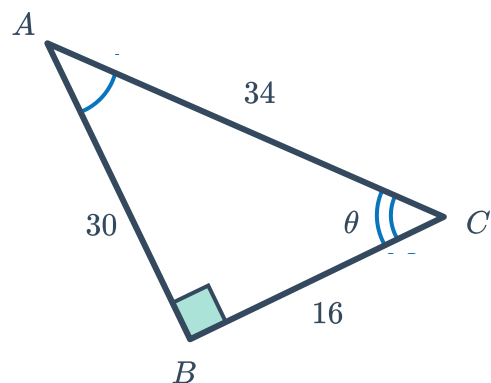
ii $\frac{\text{Opposite}}{\text{Hypotenuse}}$

iii $\frac{\text{Adjacent}}{\text{Hypotenuse}}$

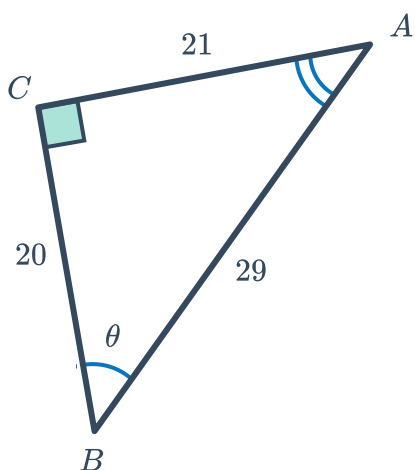
a



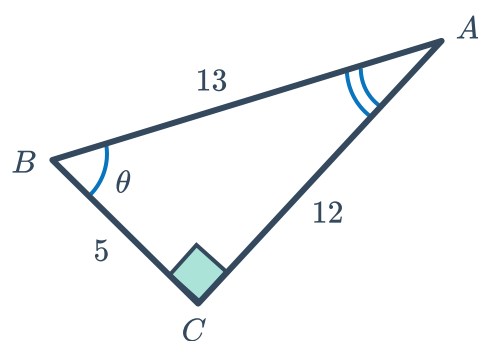
b



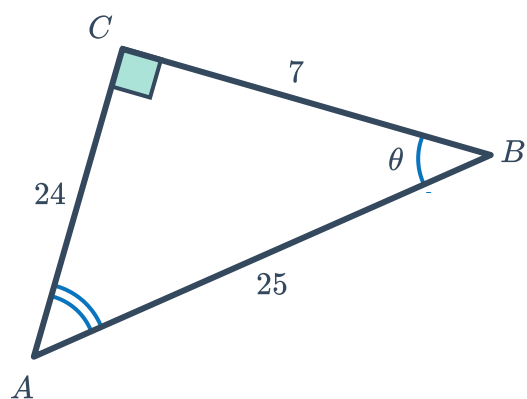
c



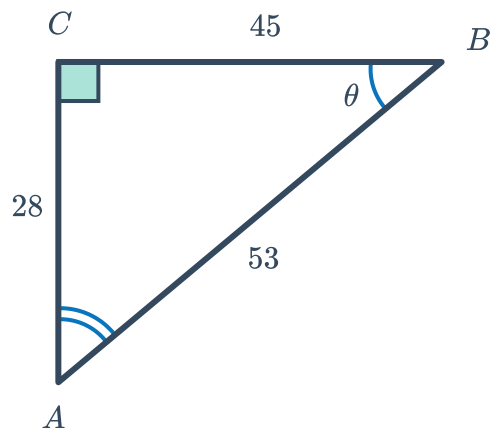
d



e



f





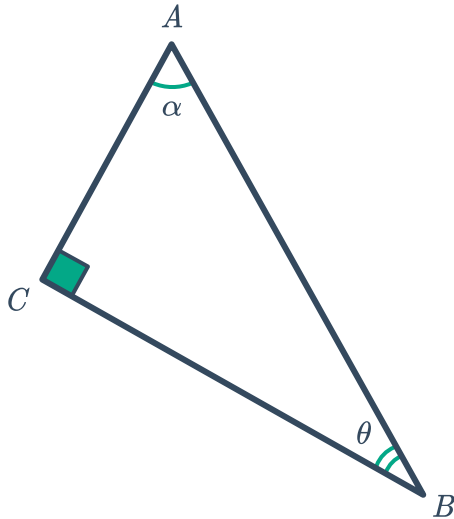
Trigonometric ratios

4 Write down the indicated ratios for the following triangles:

a

i $\tan \theta$

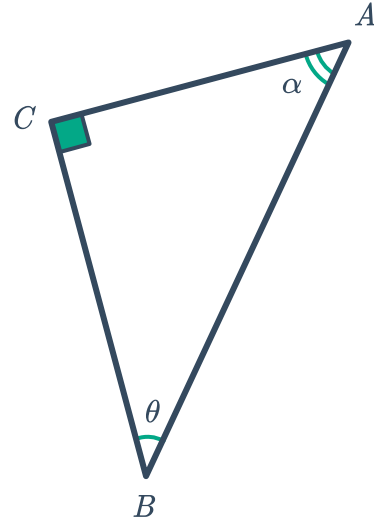
ii $\sin \alpha$



b

i $\sin \theta$

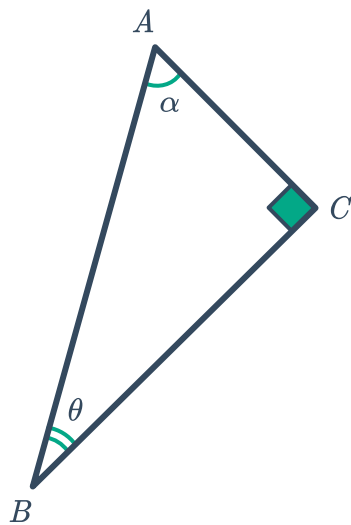
ii $\tan \alpha$



c

i $\cos \theta$

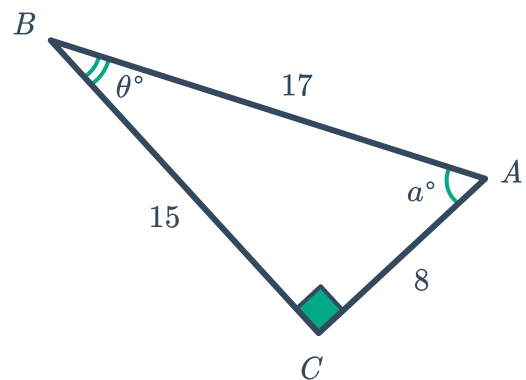
ii $\cos \alpha$



d

i $\tan \theta$

ii $\sin \alpha$



e

i $\cos \theta$

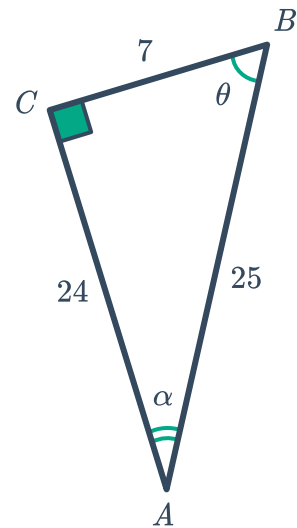
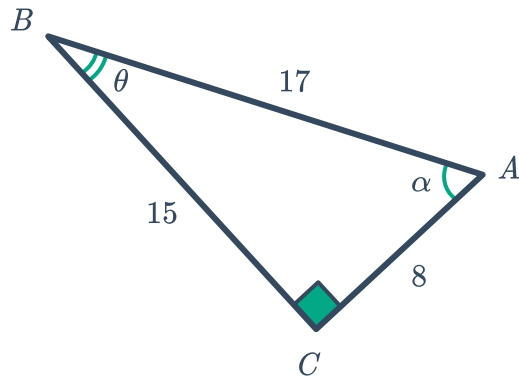
ii $\tan \alpha$



f

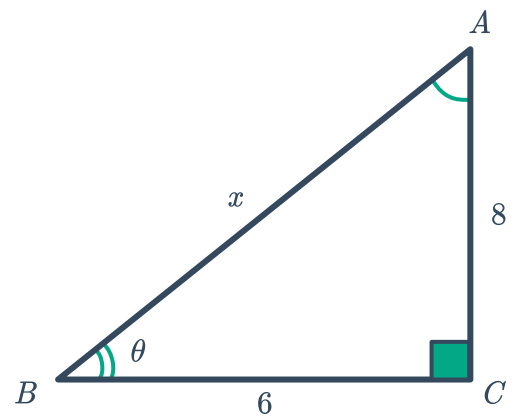
i $\sin \theta$

ii $\cos \alpha$



5 Consider the following triangle:

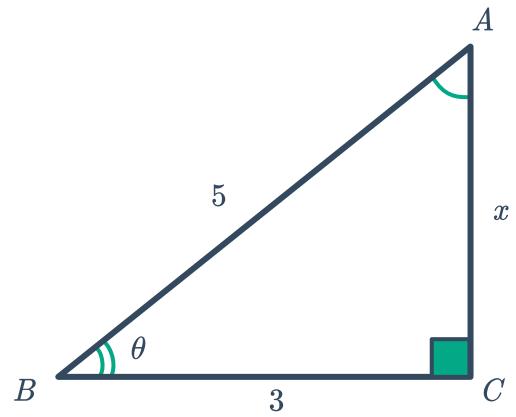
- a Find the value of x .
- b Find the value of $\sin \theta$.
- c Find the value of $\cos \theta$.



6 Consider the following triangle:

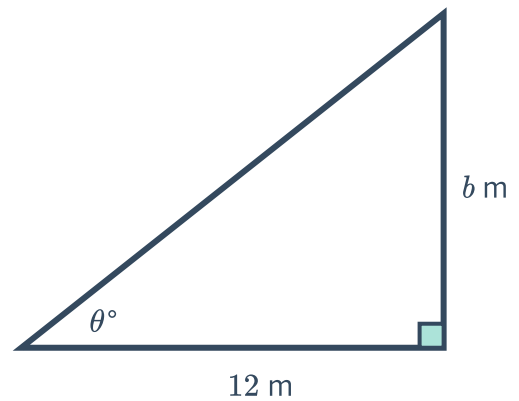


- a Find the value of x .
- b Hence, find the value of $\tan \theta$.

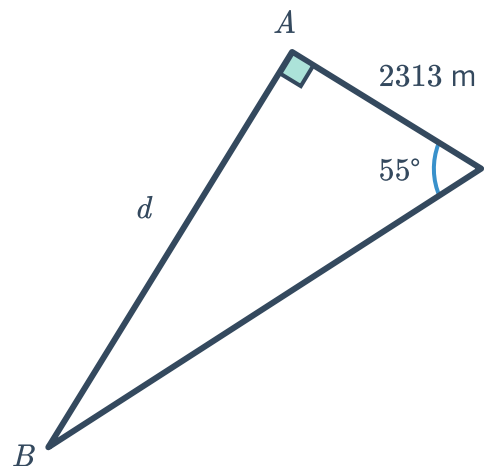


Unknown sides

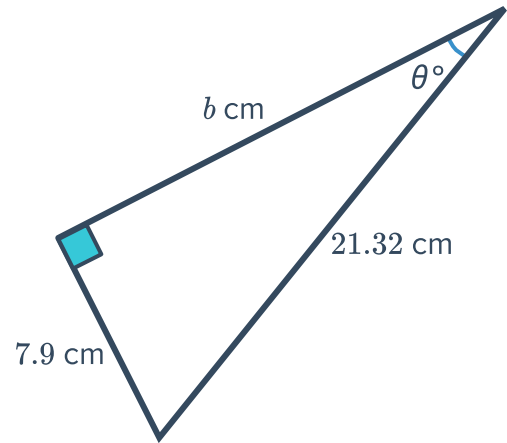
- 7 For the following triangle, if $\tan \theta = \frac{2}{3}$, find the value of b .



- 8 For the following triangle, if $\tan \theta = \frac{4}{3}$, find the value of d .

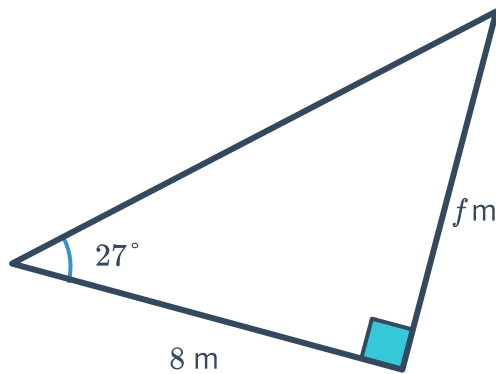


- 9 For the following triangle, if $\tan \theta = 0.4$, find the value of b .
Round your answer correct to one decimal place.

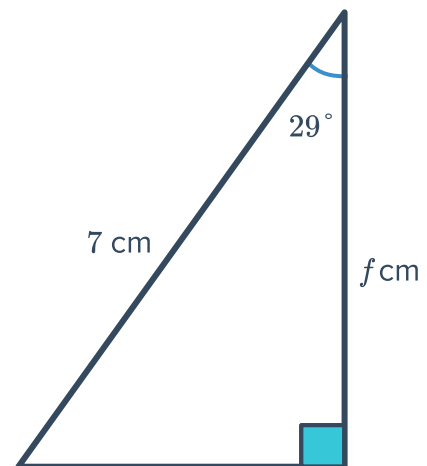


- 10 Find the value of the pronumeral in the following triangles, correct to two decimal places:

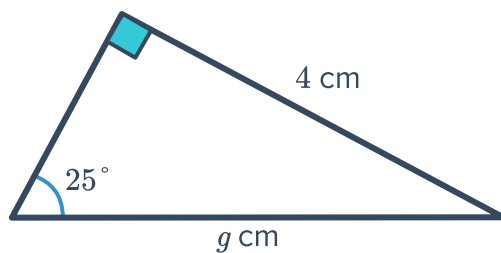
a



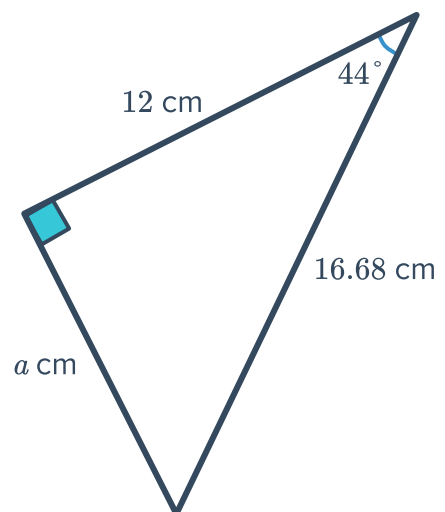
b



c



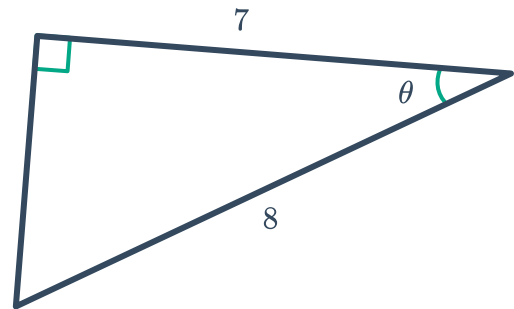
d





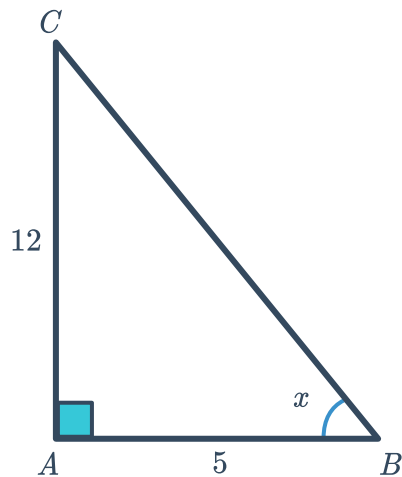
Unknown angles

- 11 Given the following triangle, calculate the exact value of $\tan \theta$.

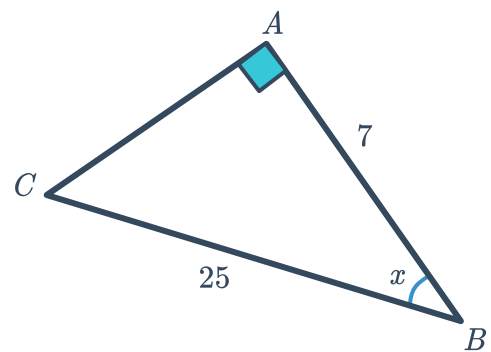


- 12 For each of the following triangles, find the value of x to the nearest degree:

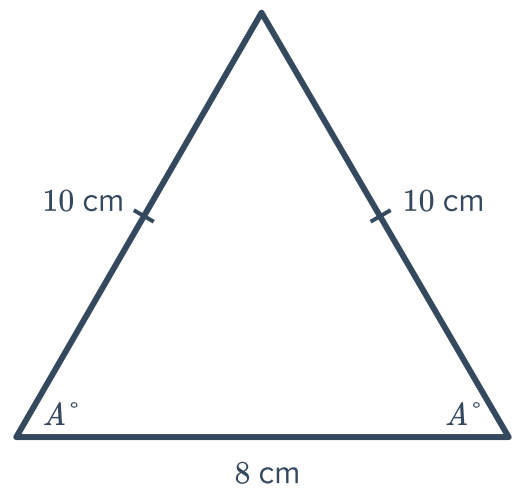
a



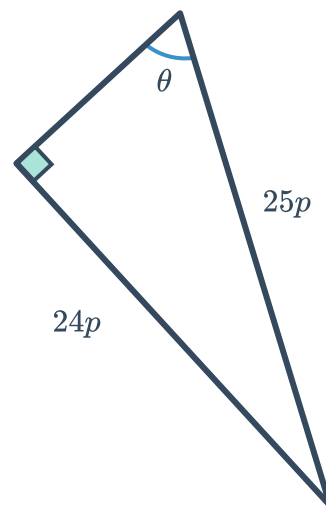
b



- 13 An isosceles triangle has equal side lengths of 10 cm and a base of 8 cm as shown. Calculate the size of angle A to one decimal place.



- 14 Find the value of $\tan \theta$ for the following triangle:

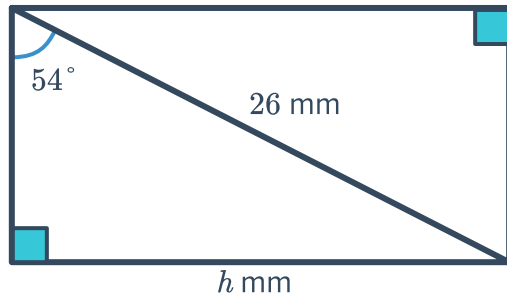




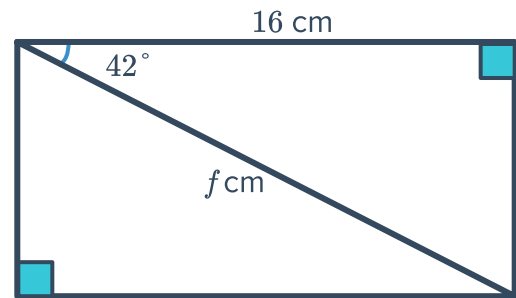
Applications

15 Find the value of the pronumeral(s) in the following diagrams, correct to the nearest whole number:

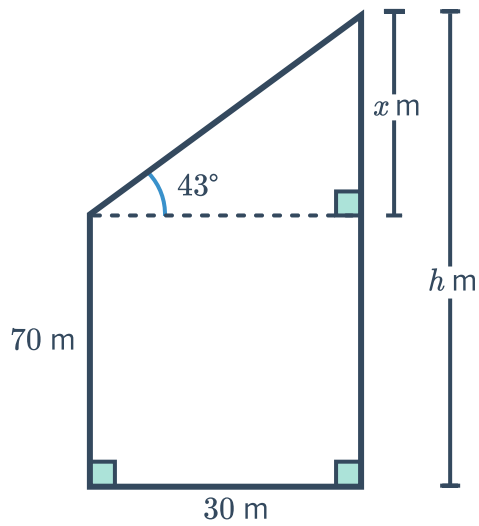
a



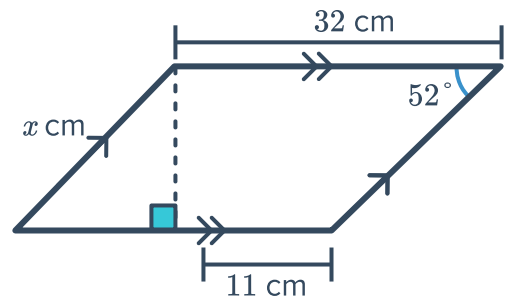
b



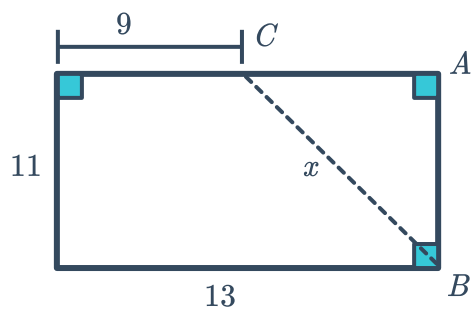
c



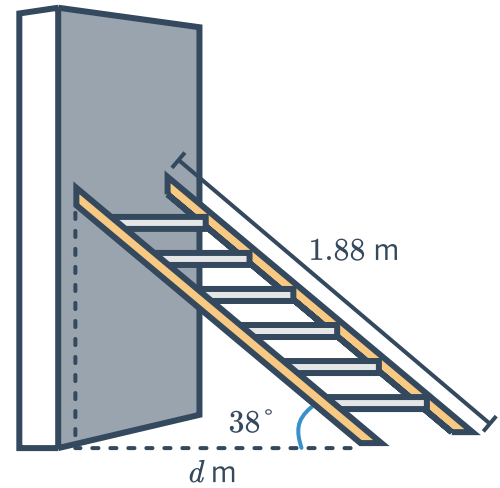
d



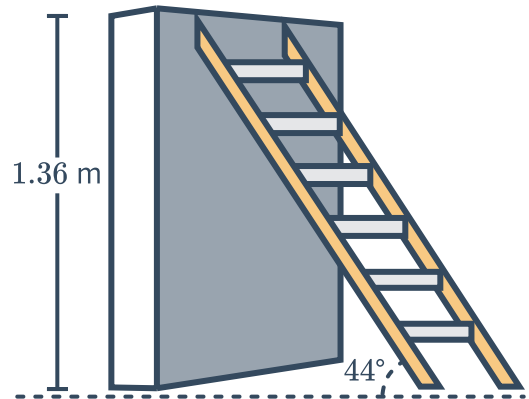
e



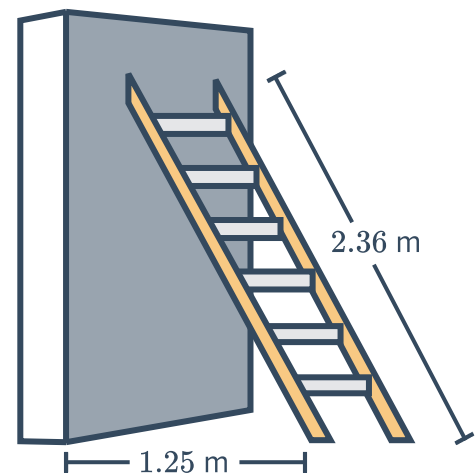
- 16 If d is the distance between the base of the wall and the base of the ladder, find the value of d to two decimal places.



- 17 A ladder is leaning at an angle of 44° against a 1.36 m high wall. Find the length of the ladder, to two decimal places.



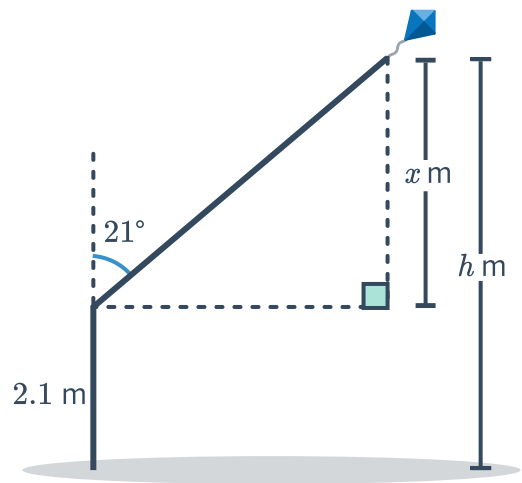
- 18 A ladder measuring 2.36 m in length is leaning against a wall. If the angle the ladder makes with the ground is y° , find the value of y to two decimal places.



- 19 A girl is flying a kite that is attached to the end of a 23.4 m length of string. The angle between the string and the vertical is 21° . The girl is holding the string 2.1 m above the ground.



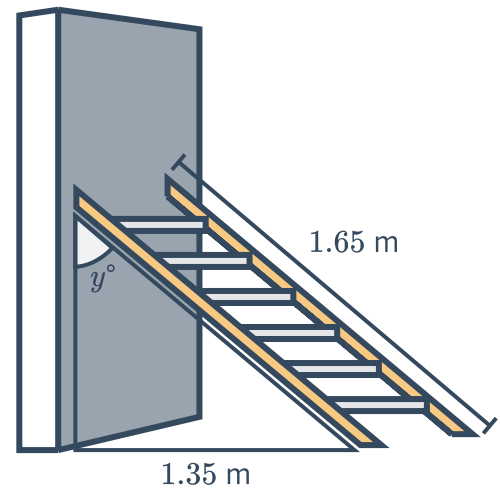
- a Find x , correct to two decimal places.
- b Hence, find the height, h , of the kite above the ground, correct to two decimal places.



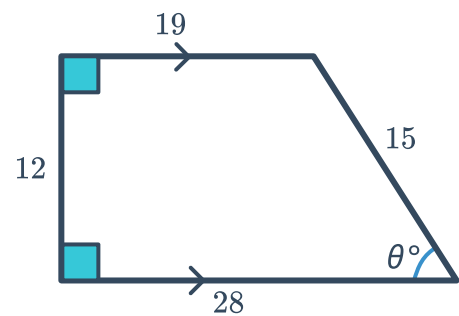
- 20 In the diagram, a string of lights joins the top of the tree to a point on the ground 23.9 m away. If the angle that the string of lights makes with the ground is θ° , find θ to two decimal places.



- 21** A ladder measuring 1.65 m in length is leaning against a wall. If the angle the ladder makes with the wall is y° , find y to two decimal places.



- 22** Find the value of $\tan \theta$ in the following trapezium:



- 23** A sand pile has an angle of 40° and is 10.6 m wide. Find the height of the sand pile, h , to one decimal place.

